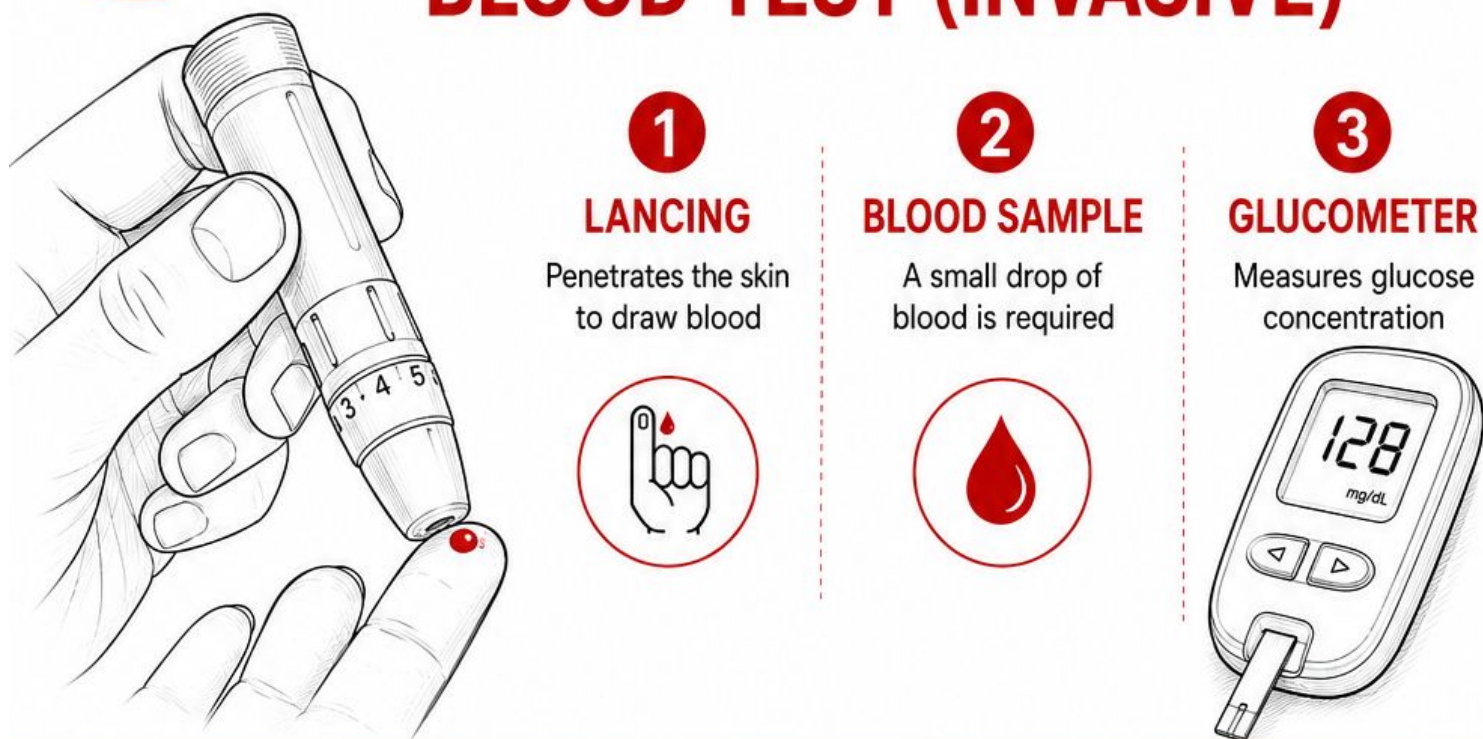


Non-Invasive Blood Glucose Level Estimation Using Smartphone PPG Signals

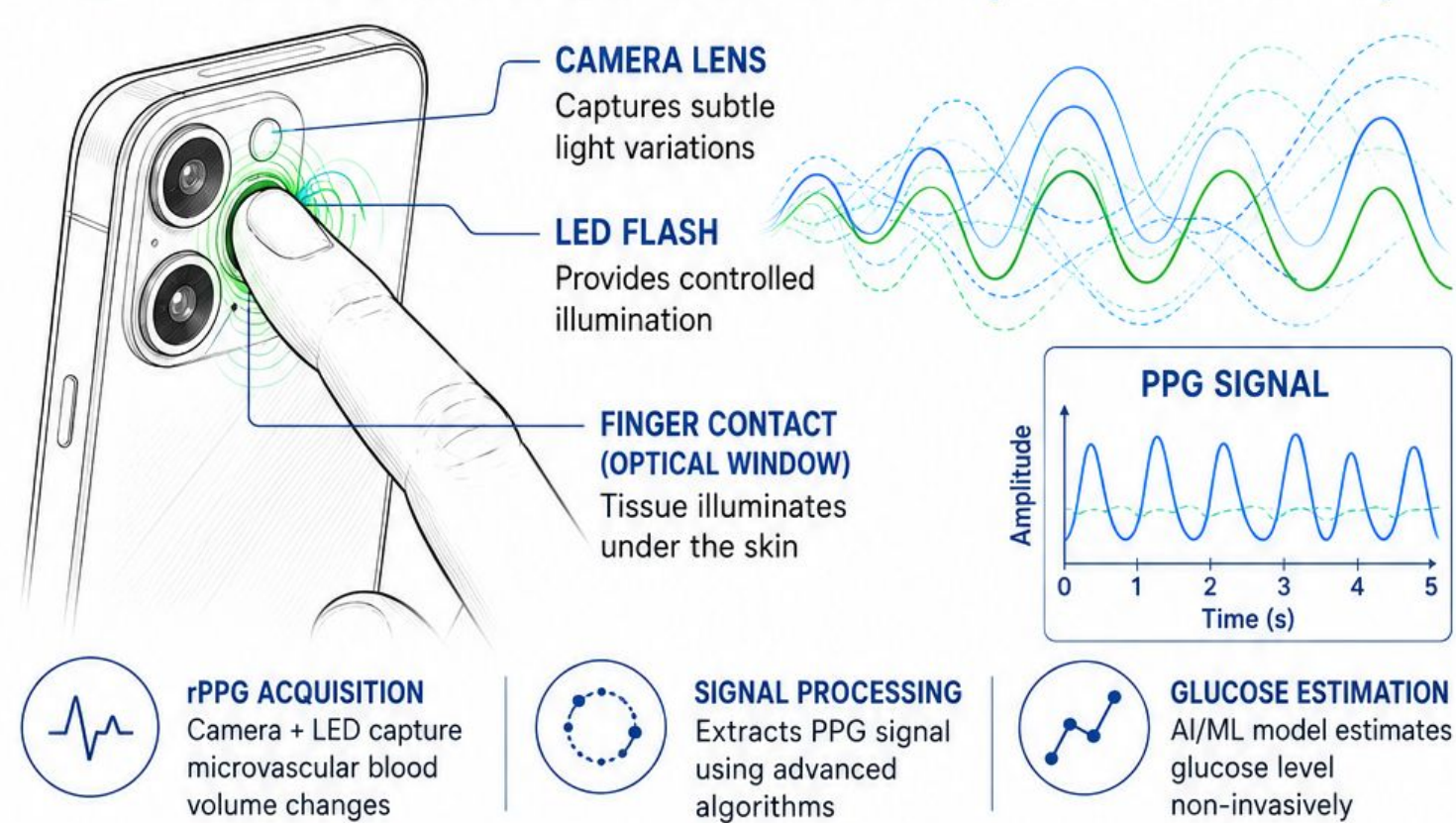
Introduction

Traditional methods for longitudinal metabolic tracking rely heavily on invasive, painful, and costly finger-prick glucometers, which continuously impose physical discomfort and financial barriers, leading to notoriously low patient compliance. To mitigate these continuous clinical challenges, this graduation project presents a novel, non-invasive alternative that leverages state-of-the-art computer vision and deep learning techniques. By transforming ubiquitous consumer-grade smartphones into decentralized medical diagnostic tools, the proposed framework captures high-fidelity remote Photoplethysmography (rPPG) signals directly from the user's fingertip using standard embedded cameras. This paradigm shift effectively eliminates the need for specialized blood-sampling hardware, paving the way for a painless, highly accessible, and proactive ecosystem for modern diabetes monitoring.

✗ TRADITIONAL FINGER-PRICK BLOOD TEST (INVASIVE)



✓ SMARTPHONE rPPG GLUCOSE ESTIMATION (NON-INVASIVE)



Evolution of Glycemic Tracking – From Invasive Finger-Prick to Non-Invasive Smartphone rPPG Estimation.

Methods and materials

The core computational framework processes raw temporal **rPPG** signals using two completely independent transfer learning pathways. To overcome the severe **overfitting** limitations observed during our initial deep **1D-ResNet** experiments, a robust multi-scale **1D-CNN** backbone was utilized instead. This backbone was first pre-trained on blood pressure estimation to effectively extract underlying **cardiovascular features**. The pre-trained weights were then frozen and fine-tuned separately across two distinct operational stages: once via a continuous **regression model** for precise glucose tracking, and once via a three-tier **classification model** (**Low** < 70, **Natural** 70–200, **High** > 200 mg/dL) engineered for targeted clinical safety alerts.

Presented by:

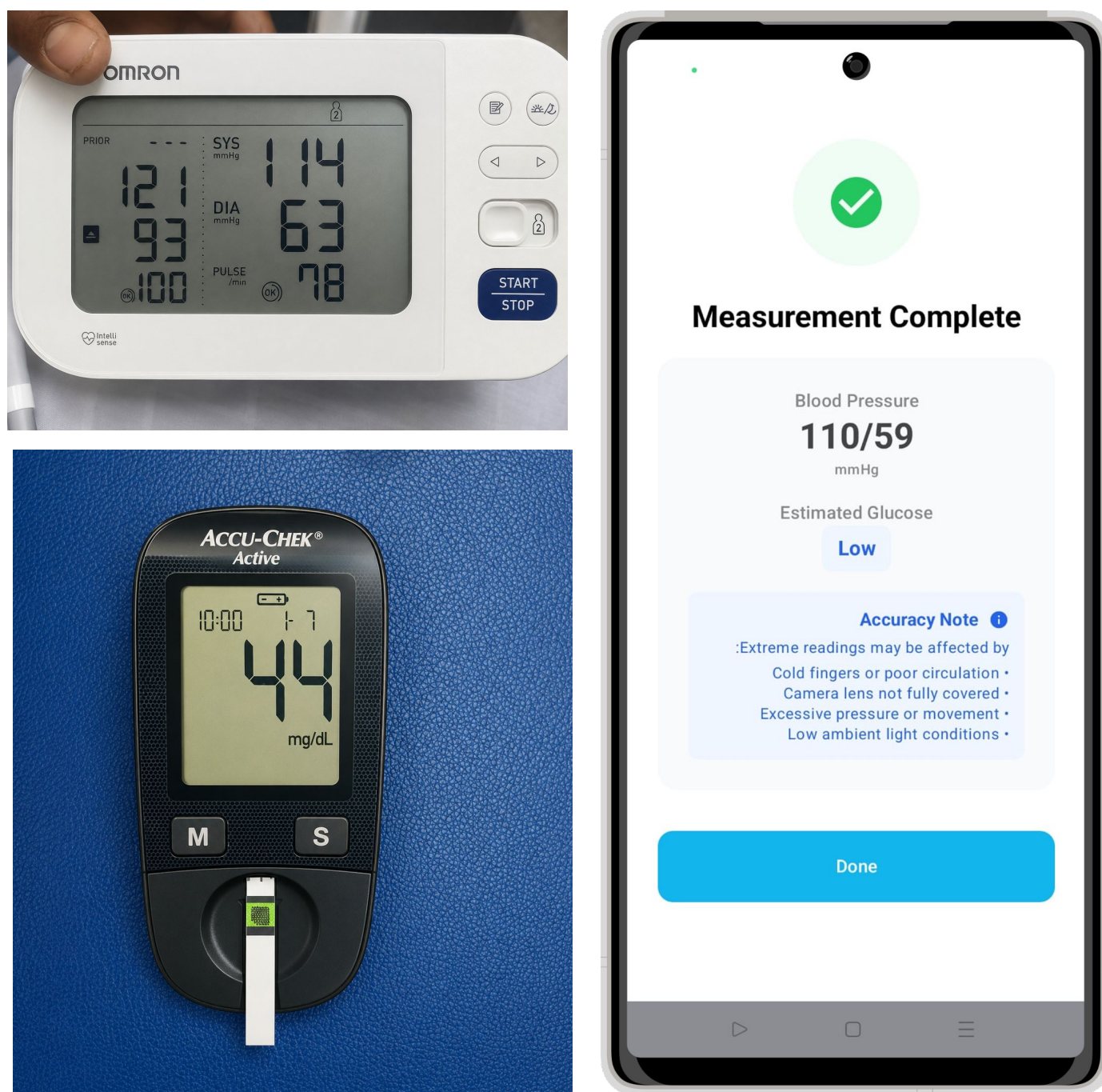
- 1) Abdelrahman Rabie Ahmed
- 2) Mostafa Magdy Ahmed
- 3) Sherif Nasser Abdellahy

System description

The end-to-end framework is fully realized through a production-ready mobile health application designed for cross-platform deployment. The smartphone software integrates an optimized runtime pipeline that manages real-time PPG signal acquisition via the mobile camera sensor while dynamically regulating the device's LED flashlight to maintain uniform tissue illumination. To ensure maximum clinical precision, the localized pipeline incorporates adaptive signal quality estimation algorithms that actively filter out environmental somatic noise, baseline wander, and minor motion artifacts. Furthermore, a minimalist biomedical UI/UX configuration delivers continuous visual feedback to guide optimal finger placement, ensuring a highly intuitive and user-friendly interaction loop.

Results

The independent transfer learning pathways demonstrated high precision and clinical viability. Real-world edge testing successfully verified the system across diverse profiles, where the mobile app precisely tracked blood pressure at **110/59 mmHg** against a commercial **114/63 mmHg** reference. Crucially, during sudden glycemic drops, the system effectively triggered built-in safety protocols, automatically masking the numerical value to display the **"Low"** classification category in perfect alignment with the **44 mg/dL** invasive glucometer reference.



Real-world testing of the dual-pathway transfer learning framework, showcasing highly aligned blood pressure tracking (110/59 vs. 114/63 mmHg) and successful deployment of the safety classification UI during a critical "Low" glycemic event (44 mg/dL reference).

Conclusions and future work

In conclusion, this project successfully bridges deep biomedical signal processing with mobile edge computing, demonstrating that consumer smartphones can be effectively repurposed as reliable, non-invasive blood glucose estimation platforms. By removing physical pain and recurring financial costs, this research establishes a scalable baseline for proactive healthcare management. Future extensions of this work will focus on launching broader clinical trials to further diversify the training demographics, exploring multi-sensor fusion paradigms by integrating contextual biometrics from commercial smartwatches, and executing deep 8-bit quantization to facilitate deployment on ultra-low-power biomedical wearables.

Supervised by:

- Prof. Osama Ahmed Omar
- Dr. Mohammed Abdelhasser Mohammed
- Dr. Ahmed Mohammed Noreldien
- Eng. Somaya Abdelkhier Mohamed